

Diagnosing and Correcting Geometric Bias in Diversity-Driven Code Instruction Data Selection

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Abstract

Diversity-driven data selection offers a scalable way to reduce code instruction-tuning costs, yet it relies on a fragile assumption: geometric spread in representation space should correspond to training-useful semantic coverage. In code data selection, our diagnosis reveals a geometric bias behind this assumption: dense embedding geometry can be strongly influenced by rare surface forms, verbose constraints, and implementation artifacts, causing diversity objectives to over-explore locally sparse or globally peripheral samples whose training utility is not guaranteed. To correct this bias, we propose *Semantically Anchored Geometric Coverage* (SAGC), a framework for reshaping representation geometry using LLM-generated macro-intent labels as semantic anchors for dense micro-context representations. Rather than replacing standard diversity objectives, SAGC reshapes the geometry exposed to them, using macro-intent anchors to constrain dense-space exploration toward high-level algorithmic structure without requiring downstream validation signals, gradient feedback, or evaluation results. Using only 20k samples selected from a 446k pool, SAGC reaches 0.4006 on LiveCodeBench, approaching full-data training at 0.4027. Further diagnostics and out-of-benchmark evaluations show that semantic anchoring provides a structured correction to this geometric bias.

CCS Concepts

• **Computing methodologies** → **Machine learning**; **Natural language processing**; • **Information systems** → **Data mining**.

Keywords

data selection, code instruction tuning, diversity-driven sampling

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1 Introduction

Supervised fine-tuning (SFT) is fundamental for adapting large language models (LLMs) to domain-specific requirements [21, 32]. As instruction corpora grow to million-scale, especially for code SFT [1], full-data training becomes increasingly expensive, motivating compact subset selection. Among data selection strategies, diversity-driven selection is particularly attractive because it operates directly on the candidate pool without validation data, task-specific feedback, or gradient-based influence estimates [4]. Common objectives include aggregate distance maximization (Max-Sum) [10], space-filling coverage (Max-Min) [11], and clustering-based representative selection (e.g., K-Means) [20, 23], which respectively favor globally separated samples, coverage of uncovered regions, and cluster-based representatives. Despite these differences, they all depend on the geometry of the representation space. *Their effectiveness shares a critical assumption: the geometric signals they optimize—distances, densities, or uncovered regions—must correspond to semantic variation useful for fine-tuning.*

This assumption is particularly fragile in code instruction tuning. Code instructions often exhibit multiple layers of variation, including surface-level wording and constraints, implementation-level context, and high-level algorithmic intent [7, 24, 27]. Dense representations can encode fine-grained contextual cues such as phrasing, constraints, control flow, and implementation patterns, but they do not reliably indicate whether a large geometric distance reflects functional novelty or merely surface-form rarity. In long-tailed code instruction data, where code and programming artifacts often exhibit highly skewed distributions [2, 13], rare lexical patterns, verbose constraints, unusual problem formulations, and idiosyncratic coding artifacts can therefore appear overly novel to distance-based objectives. This creates a risk of geometric bias: dense-space diversity may over-prioritize samples that appear geometrically distinctive because they are rare, locally isolated, or globally peripheral, rather than because they provide training-useful semantic coverage. *We therefore ask whether standard diversity objectives over dense code representations expand useful coverage, or instead amplify rare, locally isolated, or globally peripheral regions of the candidate pool.* This question clarifies how dense representation geometry shapes standard diversity objectives, motivating our focus on representation-level correction.

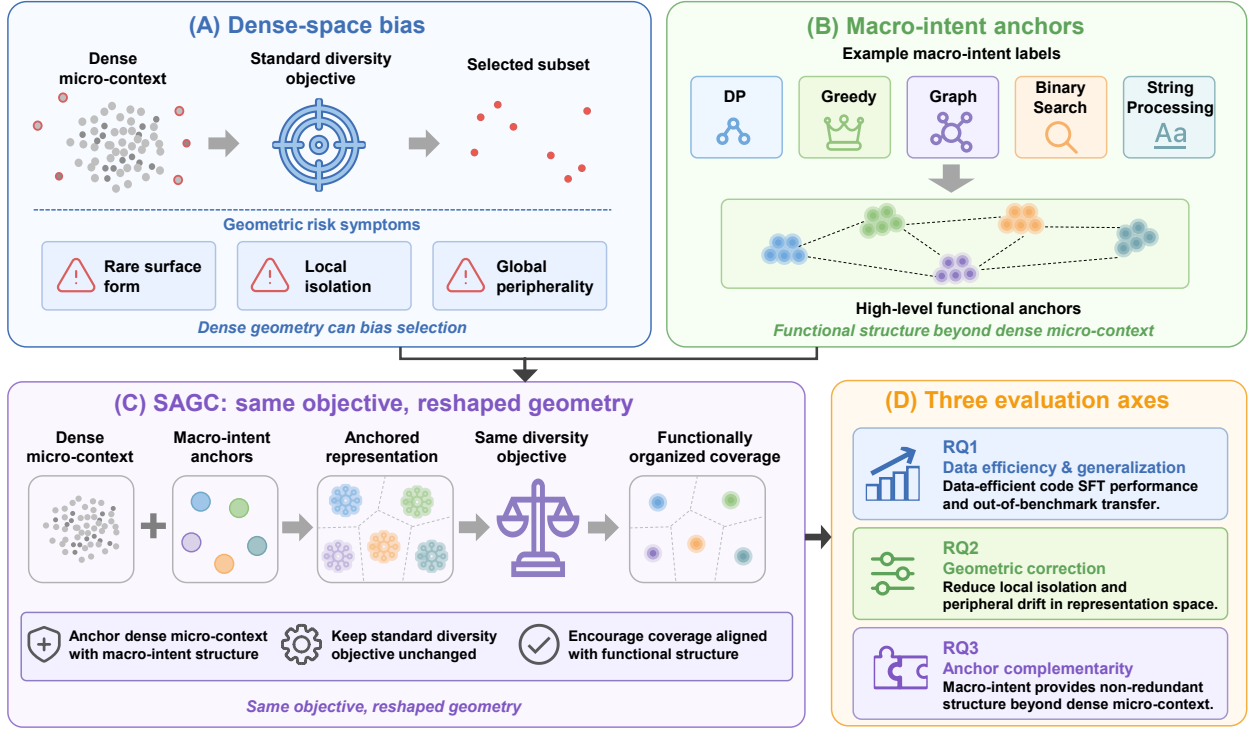


Figure 1: Overview of SAGC. (A) Dense-space diversity selection can favor samples that are rare, locally isolated, or globally peripheral. (B) Macro-intent labels provide high-level functional anchors beyond dense micro-context. (C) SAGC anchors dense micro-context with macro-intent structure, keeping the diversity objective unchanged while reshaping the geometry exposed to selection. (D) We evaluate SAGC along three axes: data-efficient code SFT and generalization, geometric correction, and anchor complementarity.

We first diagnose geometric bias by characterizing how standard diversity objectives behave under dense code representations. On a 446k-sample code instruction pool, we evaluate dense-only selectors along surface rarity, intent distribution, embedding-space atypicality, and downstream utility. The results show that distance-based dense selection can substantially amplify surface-level rarity and geometric atypicality. For example, applying Max-Min to dense code embeddings increases the rare instruction-token ratio from 0.10% under random sampling to 0.59%. It also increases the fraction of samples falling in locally sparse neighborhoods from 2.23% to 40.34%, and the fraction of samples near the global periphery of the embedding space from 0.68% to 6.16%. Yet this aggressive geometric exploration does not translate into better downstream utility: the corresponding dense code Max-Min selector performs worse than the strong random baseline on LiveCodeBench. These results reveal a geometric bias in dense-space diversity selection: standard objectives can reshape the selected subset, but under dense code geometry they may do so in directions dominated by rare surface forms, local isolation, and global peripherality rather than training-useful variation. *This diagnosis suggests that improving diversity-driven selection does not necessarily require a new diversity objective; instead, it requires changing the geometry exposed to existing objectives.*

Motivated by this diagnosis, we then correct this geometric bias through semantic anchoring. Rather than discarding dense geometry or replacing it with coarse intent labels, we seek to make geometric diversity more intent-aware. We propose **Semantically Anchored Geometric Coverage (SAGC)**, a representation-level correction for diversity-driven code instruction data selection. SAGC keeps standard diversity objectives unchanged, but reshapes the selection geometry by augmenting dense micro-context representations with LLM-generated macro-intent anchors. This anchored geometry makes standard objectives less dominated by dense-space artifacts and more aligned with high-level functional structure. Macro-intent anchors and dense micro-context therefore play complementary roles: the former provides functional organization that is often invisible to dense surface-level geometry, while the latter preserves implementation-level distinctions that label-only selection may collapse. As a result, under the semantically anchored representation geometry, the same diversity objectives can prioritize functionally organized coverage rather than unconstrained dense-space novelty, thereby supporting data-efficient code SFT.

Finally, we evaluate whether SAGC enables data-efficient code SFT and generalizes beyond the main benchmark. Using a 20k-sample subset selected from the 446k-sample pool, SAGC

achieves an MCS of 0.4006 on LiveCodeBench [16], substantially narrowing the observed gap between a strong random baseline at 0.3823 and full-data training at 0.4027. We further examine out-of-benchmark generalization on CodeEval-Pro variants of HumanEval-Pro and MBPP-Pro [41], as well as the C++ split of Ag-LiveCodeBench-X [3], where semantically anchored selection remains competitive beyond the main LiveCodeBench evaluation. Beyond downstream scores, we analyze whether semantic anchoring produces the intended geometric correction by changing the regions selected by standard diversity objectives. We also examine representation complementarity to test whether macro-intent anchors provide non-redundant structure beyond dense micro-context. *Together, these evaluations assess SAGC along three axes: data-efficient code SFT, geometric correction, and anchor complementarity.*

Figure 1 summarizes the overall pipeline of our study, illustrating how the diagnosis of dense-space bias motivates macro-intent anchoring and how the resulting selection is evaluated.

In summary, our diagnosis, representation-level correction, and empirical study lead to the following contributions:

- We identify and characterize a representation-dependent failure mode of diversity-driven code instruction selection, showing that standard dense-space objectives can steer selected samples toward distributionally atypical regions rather than toward training-useful semantic coverage.
- We propose *Semantically Anchored Geometric Coverage (SAGC)*, a representation-level correction that anchors dense micro-context with macro-intent labels while keeping standard diversity objectives unchanged.
- We provide a comprehensive empirical study across downstream SFT, selection-behavior diagnostics, representation-complementarity analysis, and out-of-benchmark evaluation, providing evidence for the practical value of semantic anchoring for data-efficient code SFT and revealing how it affects selection geometry and representation structure.

To facilitate reproducibility, we release our code, data processing scripts, and selection metadata at <https://github.com/nju-websoft/SAGC>. The repository also includes a supplementary appendix with label-generation prompts, label statistics, implementation details, benchmark settings, and full fusion results.

2 Related Work

Diversity measurement and density-aware metrics. Diversity has been widely studied for dataset construction and subset selection. Prior work includes dataset-level diversity such as the Vendi Score [8], kernel- or set-based formulations such as DPP-style diversity [30], density-distance metrics for predicting instruction-tuning performance [40], and submodular or mixture-aware objectives that balance representativeness and diversity [28]. These methods highlight that useful diversity is not captured by distance alone: density, coverage, representativeness, and subset-level interactions all influence whether a selected set is useful for training. However, such metrics and objectives are typically defined on a pre-existing feature space, similarity kernel, or task structure. When this underlying structure emphasizes surface variation, rare lexical patterns, or peripheral samples, the resulting diversity estimate can inherit the same geometric bias. More specifically,

similarity-spectrum measures such as the Vendi Score and kernel-based methods such as DPP depend on the supplied similarity matrix; density-aware metrics such as NovelSum combine distance and density signals but still operate on the given representation space; and mixture-aware submodular methods such as SMART often require meaningful mixture or task structure to be specified in advance. *Complementary to these metric-level advances, our work does not propose another diversity score or mixture allocation rule; instead, we diagnose how standard selection objectives behave under misaligned code representations and use macro-intent anchors to recalibrate the geometry exposed to them.*

Diversity-driven data selection. A related line of work directly uses diversity to construct training subsets, including coverage-oriented procedures such as k -center-style selection [34] and large-scale representation-based frameworks such as RDS+ [15]. However, large-scale studies show that many proposed selection strategies can struggle to consistently outperform random sampling [36], suggesting that diversity-based subset construction can be sensitive to the selection setting. In instruction tuning, semantic-tag-based methods such as #InsTag [22] and MIG [6] have also used explicit semantic labels to characterize instruction diversity and guide selection. For subset selection, an important question is not only whether a rule increases diversity, but which regions of the candidate pool it selects. When the representation space overemphasizes surface or peripheral variation, diversity-driven selection may favor samples that are geometrically distinct but not necessarily aligned with training utility. *Our work studies this representation-dependent sensitivity directly; SAGC does not replace standard objectives with a tag-space selection rule, but recalibrates the geometry on which standard diversity objectives operate through macro-intent anchoring.*

Utility-aware and hybrid data selection. Another line of work estimates sample utility through quality, difficulty, uncertainty, or training dynamics. Representative methods include probability- or self-scoring approaches such as IFD [18], SelectIT [19], T-SHIRT [9], and D3 [42], and training-dynamics methods such as MoSo [29] and LESS [35]. Hybrid approaches further combine multiple signals or operations, as in GraphFilter [33], MergeIT [5], and DataMaster [43]. These methods are effective when additional utility signals, graph structures, or synthesis operations are available. *In contrast, our study does not estimate sample utility, optimize graph-based quality-diversity objectives, synthesize new instructions, or rely on downstream validation signals or gradient feedback; instead, it asks how representation geometry shapes diversity-based selection. SAGC is therefore orthogonal to utility-aware and hybrid selection, providing a representation-level correction based on macro-intent anchors that can complement quality, difficulty, graph-based, or influence-based signals.*

Representations for code data selection. Large and heterogeneous code instruction corpora such as OpenCodeInstruct [1] make selection representations increasingly important. Compared with general instruction data, code examples exhibit multiple semantic levels, including prompt formulation, implementation details, language-specific patterns, and higher-level algorithmic intent [7, 24, 27]. Recent representation-oriented methods use model-native hidden states [15] or structured latent signals such as sparse autoencoder features [39], demonstrating that the representation used

for selection is crucial. However, these approaches primarily derive the selection geometry from dense, hidden-state, or activation-based views of the data. Such representations can capture rich local semantic and implementation-level cues, but do not necessarily explicitly organize higher-level functional intent that diversity objectives should cover. This distinction is important in code SFT, where samples that differ substantially in surface form or implementation details may still exercise similar algorithmic intent. *SAGC addresses this by using macro-intent labels as semantic anchors for dense micro-context, so that standard diversity objectives explore fine-grained variation under an explicit functional organization.*

3 Problem Formulation

Given a code generation instruction pool $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ and a target subset size $B \ll N$, our goal is to select a subset $S \subseteq \mathcal{D}$ with $|S| = B$ that preserves the downstream coding capability of an LLM fine-tuned on S . Ideally, the optimal subset is defined as

$$S^* = \arg \max_{S \subseteq \mathcal{D}, |S|=B} \mathcal{P}(S), \quad (1)$$

where $\mathcal{P}(S)$ denotes the downstream performance of the model after SFT on S .

The SFT process optimizes model parameters θ by minimizing the generation loss over the selected subset:

$$\theta_S^* = \arg \min_{\theta} \sum_{(x,y) \in S} \mathcal{L}_{\text{gen}}(x, y; \theta), \quad (2)$$

where $\mathcal{L}_{\text{gen}}$ is the standard next-token prediction loss. The resulting performance is evaluated on an independent evaluation set $\mathcal{D}_{\text{eval}}$:

$$\mathcal{P}(S) = \mathbb{E}_{(x_{\text{eval}}, y_{\text{eval}}) \sim \mathcal{D}_{\text{eval}}} [\text{Metric}(f_{\theta_S^*}(x_{\text{eval}}), y_{\text{eval}})], \quad (3)$$

where $f_{\theta_S^*}(\cdot)$ denotes the fine-tuned model and $\text{Metric}(\cdot, \cdot)$ measures task performance, such as functional correctness against test cases.

Directly optimizing Eq. (1) is computationally infeasible, as it would require repeatedly fine-tuning and evaluating models over many candidate subsets. Diversity-driven selection therefore replaces the downstream objective $\mathcal{P}(S)$ with a tractable proxy based on a representation Φ and a diversity objective $\mathcal{H}$:

$$\hat{S} = \arg \max_{S \subseteq \mathcal{D}, |S|=B} \mathcal{H}(S; \Phi). \quad (4)$$

Here, $\mathcal{H}$ specifies how diversity is operationalized, while Φ determines the representation geometry exposed to the selector. This decomposition is central to our study: even under the same diversity objective $\mathcal{H}$, the selected subset $\hat{S}$ can change substantially depending on whether the representation geometry Φ reflects training-useful semantic variation or distributional artifacts. The next section diagnoses this objective–representation interaction.

4 Motivating Diagnosis: When Geometric Diversity Misleads

Building on this formulation, the reliability of diversity-driven selection becomes a question about geometry: when a selector optimizes distances, densities, or uncovered regions in a representation space, do these geometric signals actually correspond to training-useful semantic variation? If the representation geometry is biased,

the same diversity objective may favor samples that are geometrically atypical but not necessarily beneficial for fine-tuning. This issue is especially relevant for code instruction tuning, where rare surface forms, verbose constraints, unusual problem formulations, and implementation artifacts can make samples appear more novel than their training utility warrants. We therefore diagnose how representative geometric diversity objectives behave under different representation spaces, focusing on whether they expand useful coverage or drift toward lexically rare, locally isolated, or globally peripheral regions.

4.1 Geometric Diversity Objectives

Different objectives $\mathcal{H}$ induce distinct geometric tendencies over the representation space. We analyze three representative classes of diversity objectives that are widely used in data selection.

Aggregate Distance Maximization. Objectives in this category favor globally separated samples by maximizing aggregate pairwise distances. A representative example is **Max-Sum** [10], which iteratively selects samples that contribute the largest total distance to the current subset. This objective naturally encourages exploration toward geometrically distant and potentially peripheral regions of the embedding space. Its behavior therefore depends on what large distances represent in the chosen representation space. **Coverage-Driven Space Filling.** Coverage-driven objectives aim to reduce uncovered regions in the representation space. A representative example is **Max-Min** [11], which iteratively selects the sample with the largest minimum distance to the current subset. This strategy expands geometric coverage by prioritizing regions that are far from already selected samples. Its behavior therefore depends on whether such uncovered regions correspond to meaningful semantic gaps or merely locally sparse regions of the embedding space.

Density-Driven Representative Selection. Density-driven objectives prioritize representative coverage of high-density regions. A representative example is **K-Means** [20, 23], which partitions the embedding space into clusters and selects samples from these clusters under an approximately balanced allocation. Compared with distance-based objectives, such methods are typically more conservative because the cluster structure is shaped by high-density regions of the data distribution. Their behavior therefore depends on whether dense regions of the representation space capture the semantic variation needed for fine-tuning.

These objectives reflect different geometric assumptions about what constitutes useful diversity. Our goal is not to introduce new objectives, but to examine how these geometric tendencies shape the selected subsets and whether they align with training-useful coverage in code instruction tuning.

4.2 Diagnostic Metrics

To characterize their effects, we analyze selected subsets along four dimensions: surface rarity, intent distribution, embedding-space atypicality, and downstream utility. We compute these diagnostics after selection on 20k-sample subsets drawn from a 446k-sample code instruction pool; full dataset construction, SFT settings, and evaluation protocols are provided in Section 6.1. Unless otherwise specified, embedding-space diagnostics are computed on ℓ_2 -normalized

dense code embeddings, which serve as a common reference for comparing different selectors. Additional diagnostic implementation details are provided in the repository appendix.

Surface rarity. We report the rare instruction-token ratio R_{rare} , defined as the average fraction of instruction tokens whose frequency in the full pool is no larger than $\eta = 5$. Higher values indicate that subsets contain more low-frequency surface forms.

Intent distribution. To analyze the high-level functional composition of selected subsets, we use macro-intent labels as diagnostic metadata rather than as inputs to the dense-only diagnosis. These labels are normalized algorithmic tags that summarize high-level functional intents, such as dynamic programming, graph traversal, binary search, or prefix sums. We report label entropy H and top-10 label mass M_{10} over labels appearing in the selected subset. Higher entropy and lower top-10 mass indicate broader apparent intent coverage.

Embedding-space atypicality. We use two geometric diagnostics. Let $\phi(x)$ denote the ℓ_2 -normalized dense code embedding of sample x .

Local isolation measures whether selected samples lie in locally sparse neighborhoods. For sample x , let $\mathcal{N}_k(x)$ denote its k nearest neighbors in $\mathcal{D} \setminus \{x\}$ under squared Euclidean distance. We define Relative Isolation (RI) as

$$\text{RI}(x) = \frac{\frac{1}{k} \sum_{x' \in \mathcal{N}_k(x)} \|\phi(x) - \phi(x')\|_2^2}{\text{Avg}_{u,v \in \mathcal{D}, u \neq v} \|\phi(u) - \phi(v)\|_2^2}. \quad (5)$$

Global dispersion measures whether selected samples lie near the global periphery. Let μ_{dense} be the centroid of the normalized dense embeddings in $\mathcal{D}$. We define Relative Centroid Distance (RCD) as

$$\text{RCD}(x) = \frac{\|\phi(x) - \mu_{\text{dense}}\|_2}{\frac{1}{|\mathcal{D}|} \sum_{x' \in \mathcal{D}} \|\phi(x') - \mu_{\text{dense}}\|_2}. \quad (6)$$

For compact reporting, we use r_{RI} and r_{RCD} to denote the fractions of selected samples whose RI or RCD exceeds the corresponding random-baseline threshold $\mu_{\text{rand}} + 3\sigma_{\text{rand}}$.

Downstream utility. We use LiveCodeBench performance as a downstream utility indicator. We report Mean Case Success (MCS), which measures the average proportion of passed test cases per problem, following the full evaluation protocol in Section 6.1. This tests whether stronger geometric exploration aligns with fine-tuning utility.

4.3 Utility and Geometric Risk under Dense Diversity

Table 1 summarizes the behavior of dense-only diversity selectors. We include random sampling as a distributional reference, with its MCS averaged over five seeds, and compare three objectives under two dense representations. *Prompt* uses the instruction x as the embedding input, while *Code* uses the code portion extracted from the response y . Both dense representations use Qwen3-Embedding-8B [26] as the encoder. For readability, we refer to each configuration as *Embedding* / *Objective* in the following discussion.

A salient pattern is that dense diversity does not consistently outperform random sampling. Against the random baseline of 0.3823 MCS, most dense-only selectors are comparable or worse: *Prompt* / *K-Means* reaches 0.3822, *Prompt* / *Max-Min* reaches

Table 1: Dense-only diagnostic summary. Metrics follow the four diagnostic dimensions in Section 4.2: surface rarity (R_{rare}), intent distribution (H and M_{10}), embedding-space atypicality (r_{RI} and r_{RCD}), and downstream utility (MCS).

		Surface	Intent		Embed. Atyp.		Utility
		R_{rare} (%)	H	M_{10} (%)	r_{RI} (%)	r_{RCD} (%)	MCS
Random Sampling		0.10	5.39	34.25	2.23	0.68	0.3823
Objective	Embed.						
Max-Sum	Prompt	0.35	5.92	23.93	20.08	9.45	0.3476
Max-Sum	Code	0.29	5.97	23.09	21.31	14.69	0.3808
Max-Min	Prompt	0.66	5.99	23.75	32.14	5.08	0.3818
Max-Min	Code	0.59	5.93	25.87	40.34	6.16	0.3616
K-Means	Prompt	0.11	5.45	33.22	2.34	0.84	0.3822
K-Means	Code	0.12	5.46	33.25	2.86	0.85	0.3844

0.3818, and *Code* / *Max-Sum* reaches 0.3808, while *Prompt* / *Max-Sum* and *Code* / *Max-Min* drop to 0.3476 and 0.3616. Only *Code* / *K-Means* provides a slight improvement, reaching 0.3844. This pattern is consistent with prior observations that diversity-based selection can be difficult to improve over random sampling [36]. It also suggests that the selected subsets require further diagnosis: the issue may lie not only in how much the selector explores, but also in what regions of the dense space are being explored.

Beyond downstream utility, the diagnostic columns reveal how different objectives shape the selected subsets. For a fixed objective, the non-utility profiles of Prompt and Code are often similar: Max-Sum consistently increases embedding-space atypicality, while Max-Min consistently increases surface rarity and local isolation. *This suggests that the failure mode arises from the interaction between standard diversity objectives and dense representation geometry rather than from a single input view.*

K-Means largely exhibits distribution-preserving behavior. Across both prompt and code embeddings, its diagnostic profile remains close to random sampling: R_{rare} changes only from 0.10% to 0.11–0.12%, r_{RI} remains low at 2.34–2.86%, and MCS stays near the random baseline at 0.3822–0.3844. This suggests that, under dense prompt/code representations, K-Means mainly preserves the pool-level distribution rather than substantially reshaping the selected subset. Thus, while K-Means avoids the extreme geometric drift of distance-based objectives, it provides limited evidence of expanding training-useful coverage beyond random sampling.

Distance-based objectives alter the selected subsets more substantially, but the changes are not purely beneficial. Compared with random sampling, Max-Sum and Max-Min substantially broaden apparent intent coverage: entropy rises from 5.39 to 5.92–5.99 under Prompt and 5.93–5.97 under Code, while top-10 label mass drops from 34.25% to 23.09–25.87%. However, this intent-level broadening arises without explicitly reasoning about macro-intent, and it is accompanied by clear increases in surface rarity and embedding-space atypicality.

The two distance-based objectives reveal different forms of misaligned geometric exploration. **Max-Sum disproportionately pushes**

selections toward the global periphery: r_{RCD} reaches 9.45% under Prompt and 14.69% under Code, far above the random baseline of 0.68%. **Max-Min, in contrast, over-expands into locally isolated regions:** r_{RI} rises from 2.23% under random sampling to 32.14% under Prompt and 40.34% under Code. Max-Min also shows the strongest increase in surface rarity, with R_{rare} rising from 0.10% to 0.59–0.66%. *This suggests that dense-space exploration often prioritizes low-level distributional outliers, such as lexically rare, locally sparse, or globally peripheral samples, rather than directions that consistently translate into downstream utility.*

These geometric shifts help explain the weak utility gains in Table 1. Although distance-based objectives appear more diverse by both intent-distribution and embedding-space measures, this increased diversity does not consistently improve MCS. *Thus, the failure mode is not simply that dense diversity fails to explore; rather, it explores in directions that are only weakly aligned with fine-tuning utility.*

Taken together, Table 1 suggests two ways in which dense geometric diversity can fail as a proxy for useful coverage. Density-driven selection remains close to the pool distribution, while distance-based selection more aggressively shifts toward lexically rare, locally isolated, or globally peripheral samples. Neither behavior consistently improves fine-tuning utility over random sampling. *This motivates a representation-level correction: the selector should not merely choose between conservative density following and aggressive dense-space exploration, but should operate on a geometry constrained by higher-level algorithmic intent.*

5 Semantically Anchored Geometric Coverage

Building on the diagnosis in Section 4, we focus on the representation geometry exposed to diversity objectives. Rather than replacing existing diversity objectives, we reshape the representation space on which they operate. We propose **Semantically Anchored Geometric Coverage** (SAGC), a representation-level correction for diversity-driven code instruction data selection. SAGC augments dense micro-context representations with instruction-level macro-intent anchors. The resulting anchored representation is designed to guide standard diversity objectives toward functionally organized coverage while preserving fine-grained implementation context. SAGC therefore aims to support data-efficient code SFT by guiding subset selection without requiring downstream validation signals or gradient-based feedback.

5.1 Micro-Context and Macro-Intent Representations

Micro-context representation. For each instruction-response pair $(x, y) \in \mathcal{D}$, we construct a dense micro-context representation from either the instruction or the code portion of the response. Let $c(y)$ denote the code portion extracted from the response y . Given a dense encoder $\mathcal{M}_{\text{dense}}$, we encode an input view $u \in \{x, c(y)\}$ as:

$$\mathbf{v}_{\text{micro}}(u) = \text{Normalize}(\mathcal{M}_{\text{dense}}(u)). \quad (7)$$

where $\text{Normalize}(\cdot)$ denotes ℓ_2 normalization. The instruction view captures problem statements, constraints, and natural-language formulation, while the code view captures implementation details such as control flow, data structures, library usage, and coding patterns.

These dense representations provide fine-grained contextual information and thus preserve implementation-level distinctions, but, as diagnosed earlier, their geometry alone may not reliably distinguish training-useful semantic novelty from surface or implementation-level atypicality.

Macro-intent representation. To introduce higher-level functional structure beyond dense micro-context, we generate macro-intent labels for each instruction. Macro-intent labels are normalized algorithmic tags that summarize high-level properties of a coding task, such as problem goal, core algorithm, data structure, and programming paradigm. This label-generation step is performed once offline, with a modest cost relative to repeated subset optimization or SFT. It also does not require downstream validation signals, model gradients, or task execution.

For each instruction x , we prompt an LLM to generate an initial label set $\mathcal{T}_x$. We then normalize labels and apply a fixed minimum-support filter to remove extremely rare or noisy labels:

$$\mathcal{T}'_x = \{t \in \mathcal{T}_x \mid \text{freq}(t) \geq \tau\}, \quad \tau = 5. \quad (8)$$

Here, $\text{freq}(t)$ is computed over the full instruction pool. This filter is applied only to the label vocabulary, not to the data samples themselves. It removes one-off, overly specific, or spurious tags unlikely to serve as reusable semantic anchors. Long-tail samples are not discarded, and they remain represented through their dense micro-context representations and any remaining normalized labels. The resulting filtered label set $\mathcal{T}'_x$ serves as the macro-intent profile of instruction x . Label-generation prompts, normalization details, and filtering statistics are provided in the repository appendix.

To convert discrete labels into a continuous anchor, we encode each retained label t into a label embedding $\mathbf{e}_t$. Then, given the macro-intent profile $\mathcal{T}'_x$, we construct a macro-intent vector $\mathbf{v}_{\text{macro}}(x)$ using one of the following three aggregation strategies:

$$\mathbf{v}_{\text{sum}}(x) = \text{Normalize}\left(\sum_{t \in \mathcal{T}'_x} \mathbf{e}_t\right), \quad (9)$$

$$\mathbf{v}_{\text{max}}(x) = \text{Normalize}(\mathbf{z}), \quad \mathbf{z}[j] = \max_{t \in \mathcal{T}'_x} \mathbf{e}_t[j], \quad (10)$$

$$\mathbf{v}_{\text{weight}}(x) = \text{Normalize}\left(\sum_{t \in \mathcal{T}'_x} w_t \mathbf{e}_t\right), \quad w_t = \log \frac{N+1}{1 + \text{df}(t)} + 1. \quad (11)$$

Here $N = |\mathcal{D}|$ is the number of samples in the full instruction pool, and $\text{df}(t)$ denotes the number of samples whose filtered macro-intent profile contains label t .

Label-Sum captures the aggregated intent signal, *Label-Max* emphasizes dominant semantic dimensions, and *Label-Weight* up-weights more discriminative labels while down-weighting ubiquitous ones. In experiments, we instantiate macro-intent anchors with these aggregation choices and examine their effects on data-efficient code SFT, selection geometry, and micro-macro complementarity.

5.2 Anchored Geometry for Standard Diversity Selection

SAGC treats macro-intent labels as semantic anchors that reshape the geometry exposed to standard diversity objectives. In this anchored space, geometric novelty is conditioned on macro-intent structure rather than determined by dense-space distance alone, while dense micro-context preserves fine-grained implementation context. The goal is to guide subset construction toward functionally organized coverage, reducing the dominance of unconstrained dense-space novelty.

Specifically, we instantiate this anchored geometry via a parameter-free concatenation that places the normalized micro-context representation and the macro-intent representation in a shared representation space:

$$\Phi_{\text{SAGC}}(x) = \text{Normalize}(\mathbf{v}_{\text{micro}}(u) \oplus \mathbf{v}_{\text{macro}}(x)), \quad u \in \{x, c(y)\}, \quad (12)$$

where $\oplus$ denotes concatenation. Distances in the anchored space are computed using squared Euclidean distance:

$$d_{\text{SAGC}}(x, x') = \|\Phi_{\text{SAGC}}(x) - \Phi_{\text{SAGC}}(x')\|_2^2. \quad (13)$$

The key design choice is that the macro component is not used as a separate selector, a post-hoc constraint, or a label-balancing rule. Instead, it is embedded into the same representation space on which the diversity objective operates, thereby changing the neighborhood structure exposed to standard selection objectives. Under this anchored geometry, geometric novelty is evaluated jointly with high-level intent: samples that are distant mainly due to surface or implementation-level artifacts but share similar macro-intent become less dominant, whereas samples that differ in both micro-context and macro-intent remain separated. In this way, SAGC preserves fine-grained contextual variation while constraining geometric exploration with functional structure.

Returning to the proxy objective in Eq. (4), SAGC acts only on the representation term by replacing Φ with the anchored geometry Φ_{SAGC} :

$$\hat{S}_{\text{SAGC}} = \arg \max_{S \subseteq \mathcal{D}, |S|=B} \mathcal{H}(S; \Phi_{\text{SAGC}}). \quad (14)$$

In this work, $\mathcal{H}$ is instantiated with the same objective families analyzed in Section 4.1: Max-Sum, Max-Min, and K-Means. Thus, SAGC keeps the objective and optimization procedure unchanged; it makes standard diversity selection more intent-aware by changing the representation geometry on which existing objectives operate, without adding validation feedback, gradient information, or task-specific supervision.

6 Experiments

In this section, we evaluate SAGC from three complementary perspectives: data efficiency and generalization, geometric correction, and anchor complementarity. Specifically, we address the following research questions:

RQ1 (Data Efficiency & Generalization): How effective is SAGC for data-efficient code instruction selection, and does its benefit extend beyond the main evaluation setting?

RQ2 (Geometric Correction): Does semantic anchoring correct the selection behaviors predicted by our diagnosis, especially the tendency toward locally isolated or globally peripheral regions?

RQ3 (Anchor Complementarity): Does macro-intent provide non-redundant structural information beyond dense micro-context, supporting its role as a semantic anchor?

6.1 Experimental Setup

Candidate pool and selected subset size. For the candidate pool $\mathcal{D}$, we construct a large-scale code instruction pool from three instruction sources: EpiCoder-func-380k [31], Llama-Nemotron-Post-Training [25], and OpenThoughts-114k [12]. Similar to previous work [14], we filter these sources for code-related and algorithmic tasks, regenerate instruction responses with Qwen3-32B [37], and remove duplicates. The resulting pool is shared by all selection methods and contains $N = 446,062$ unique instruction-response pairs. For all main selection experiments, each method selects $B = 20,000$ samples, corresponding to approximately 4.5% of the candidate pool. This shared subset size provides a common setting for comparing representation geometries and diversity objectives. Detailed data construction steps are provided in the repository appendix.

Representations and selectors. All pool-based selectors are instantiated by combining a representation space with one of the diversity objectives analyzed in Section 4.1: Max-Sum, Max-Min, and K-Means. For dense micro-context representations, we use Qwen3-Embedding-8B (4096-dim) [26] as the primary encoder for both instruction and code views, denoted as *Prompt* and *Code*, respectively. For robustness across embedding geometries, we also use jina-code-embeddings-v1.5 (1536-dim) [17] as a code-side encoder. Macro-intent labels are generated offline using Qwen3-8B [37] and converted into Label-Sum, Label-Max, and Label-Weight representations as described in Section 5.1. SAGC variants anchor dense micro-context representations with macro-intent representations, following Section 5.2.

Comparison groups. We organize the compared methods into reference points, primary baselines, macro-only variants, generic fusion ablations, and SAGC variants. The primary baselines include *Random Sampling* and dense-only diversity selection over *Prompt* or *Code* embeddings, all using the same 20k subset size. We also report Full-Data training on the entire 446k pool as a same-setup full-pool reference for selection efficiency. Macro-only variants apply the objectives to label-derived macro-intent representations alone, assessing the utility of high-level functional structure without dense micro-context. Generic fusion ablations, including micro-micro fusion and all-feature concatenation, help assess whether the observed improvements can be explained by simply increasing feature dimensionality rather than semantic anchoring. SAGC variants correspond to the semantically anchored representations introduced in Section 5.

Evaluation benchmark and metric. Our main evaluation uses LiveCodeBench [16] problems released after 2024-09-20, which provide recent programming tasks with executable test cases and form a temporally held-out benchmark that reduces the risk of contamination. We report Mean Case Success (MCS), defined as the average proportion of passed test cases per problem. Compared with

Table 2: Main results (MCS) on LiveCodeBench. All selection methods use 20k selected samples; Full-Data uses the full 446k pool as a reference point.

Reference points			
Full-Data	0.4027		
Random Sampling	0.3823		
Representation	Diversity Objective		
	Max-Sum	K-Means	Max-Min
<i>Dense-only representations</i>			
Prompt (Qwen3)	0.3476	0.3822	0.3818
Code (Qwen3)	0.3808	0.3844	0.3616
Code (Jina)	0.3587	0.3839	0.3459
<i>Macro-only representations</i>			
Label-Sum	0.3805	0.3888	0.3949
Label-Max	0.3733	0.3882	0.3823
Label-Weight	0.3700	0.3754	0.3935
<i>Generic fusion ablations</i>			
Prompt (Qwen3) & Code (Jina)	0.3825	0.3845	0.3873
All-Concate	0.3753	0.3858	0.3943
SAGC variants			
Prompt & Label-Max	0.3768	0.4006	0.3980
Code (Qwen3) & Label-Weight	0.3822	0.3900	0.3941
Code (Jina) & Label-Weight	0.3856	0.3975	0.4002

binary Pass@1, MCS provides a more fine-grained measure of functional correctness when generated solutions pass only a subset of test cases. Additional benchmark details are provided in the repository appendix.

Training and implementation. All SFT experiments use Qwen2.5-7B [38] as the base model and train selected subsets under the same configuration; because no downstream validation signal is used, we evaluate the final checkpoint for each run. All subset construction methods operate only on the candidate pool and do not use downstream validation signals, model gradients, or evaluation feedback. For *Random Sampling*, we report the average over five independently sampled subsets. LiveCodeBench MCS is averaged over 10 independent generation runs unless noted otherwise. Training hyperparameters and selection algorithm details are provided in the repository appendix.

6.2 Main Results on LiveCodeBench

This subsection addresses the data-efficiency part of RQ1 by evaluating whether SAGC improves code instruction selection when using a small subset of the candidate pool. Table 2 reports the results with Full-Data training on the entire pool as a reference point to contextualize the performance of selected subsets. Results are organized by the representation space exposed to each diversity objective. For compactness, the main table reports representative generic fusion and SAGC configurations selected by average MCS across objectives, while the repository appendix reports the full matrix of evaluated generic fusion and SAGC variants across representations, label aggregations, and diversity objectives.

SAGC approaches full-data performance with a small selected subset. The strongest SAGC configuration, *Prompt & Label-Max* with K-Means, achieves 0.4006 MCS using only 20k selected samples, approaching full-data training at 0.4027 and outperforming a strong random baseline at 0.3823. A second SAGC configuration, *Code (Jina) & Label-Weight* with Max-Min, also reaches 0.4002, suggesting that the high selected-subset performance is not merely an isolated configuration. Together, these results answer the data-efficiency part of RQ1, showing that semantic anchoring can recover much of the observed performance gap between random selection and the same-setup full-data reference while using only about 4.5% of the candidate pool.

The representation groups further clarify that the gain depends on the representation geometry exposed to the selector. Dense-only selection largely fails to deliver consistent gains over random sampling, consistent with the diagnosis in Section 4. In contrast, macro-only representations are substantially more competitive: Label-Sum and Label-Weight reach 0.3949 and 0.3935 under Max-Min, and Label-Sum also reaches 0.3888 under K-Means. This indicates that diversity selection is not ineffective by itself; rather, its utility depends strongly on whether the representation exposes training-relevant structure to the selector. The weak behavior of dense-only selection therefore points more to the geometry of dense embeddings than to a fundamental failure of diversity objectives.

At the same time, the macro-only results show that **intent structure is useful on its own, while the strongest observed configurations arise when it complements rather than replaces dense micro-context**. This pattern suggests that macro-intent is particularly useful as an anchor for shaping the geometry in which fine-grained contextual variation is explored. The objective-wise pattern further qualifies this result: SAGC is strongest under K-Means and Max-Min, while Max-Sum yields only modest gains over random sampling even with anchoring. This is consistent with the diagnosis that aggregate distance maximization remains sensitive to peripheral geometric spread.

Generic fusion ablations further suggest that the improvement is not merely due to concatenating more features. For example, *Prompt (Qwen3) & Code (Jina)* and *All-Concate* remain below the best SAGC variants under both K-Means and Max-Min. This suggests that simply adding another dense view or increasing representation dimensionality is insufficient; the gain appears to depend on how macro-intent reshapes the selection geometry.

6.3 Geometric Correction Analysis

The main results show that semantic anchoring improves downstream utility, but they do not directly reveal how the selected subsets change. This subsection addresses RQ2 by examining whether semantic anchoring corrects the geometric behaviors identified in our diagnosis. Unlike the motivating diagnosis, which focused on failure modes of dense-only selection, this analysis asks whether the semantically anchored geometry changes the regions reached by standard diversity objectives. Using the diagnostic metrics introduced in Section 4.2, we focus on local isolation and global dispersion, which capture whether selected samples concentrate in

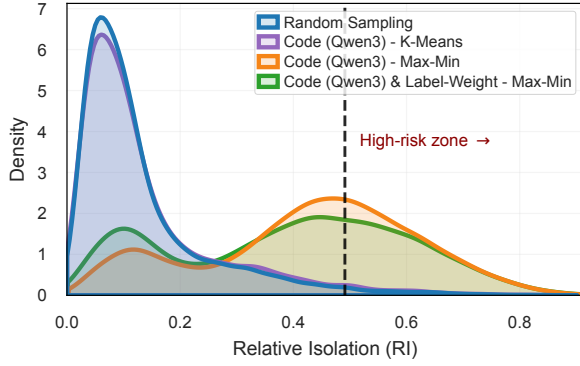


Figure 2: Local isolation distributions for representative selectors. The dashed line marks the high-risk threshold derived from the random baseline. Dense-only Max-Min places substantially more mass in the high-risk region, while the semantically anchored variant reduces this high-isolation tail.

sparse neighborhoods or drift toward the periphery of the dense space.

Figure 2 visualizes the full RI distributions behind the high-risk ratios reported in Section 4.2. **Semantic anchoring reduces the high-isolation tail induced by dense-only Max-Min while preserving non-random coverage.** The dense-only K-Means curve nearly overlaps with random sampling, confirming that its conservative behavior is distribution-level rather than limited to an aggregate statistic. In contrast, dense-only Max-Min develops a pronounced high-RI tail, showing that its coverage expansion selects a substantial fraction of locally isolated samples. The semantically anchored variant substantially reduces this tail, but its distribution remains distinct from random sampling. This indicates that the macro-intent anchor changes where Max-Min explores, rather than merely reverting the selector to a conservative baseline.

Figure 3 shows that different representation-objective pairings lead to visibly different global-dispersion patterns. **Semantic anchoring makes global dispersion more controlled, whereas adding another dense view alone does not correct peripheral drift.** For the *Prompt* - *K-Means* selected subset, the RCD distribution remains centered near the $RCD = 1$ reference line, with a slightly wider dispersion range than random sampling. Distance-based objectives shift more aggressively outward: *Prompt* - *Max-Sum* shows the most peripheral behavior, with a substantially higher median and upper range, while *Prompt* - *Max-Min* also moves away from the pool center but less extremely. In contrast, *Sum* - *Max-Min* stays close to the reference line, indicating that macro-intent structure constrains global dispersion even under a distance-based objective. The micro-micro variant, *Prompt* & *Code* - *Max-Min*, does not remove this outward shift, suggesting that adding another dense view alone is insufficient to correct peripheral drift. The semantically anchored variants show a more controlled pattern: *Prompt* & *Max* - *K-Means* stays near the reference region, and *Code* & *Weight* - *Max-Min* remains less extreme than dense distance-based selection while still maintaining non-random exploration.

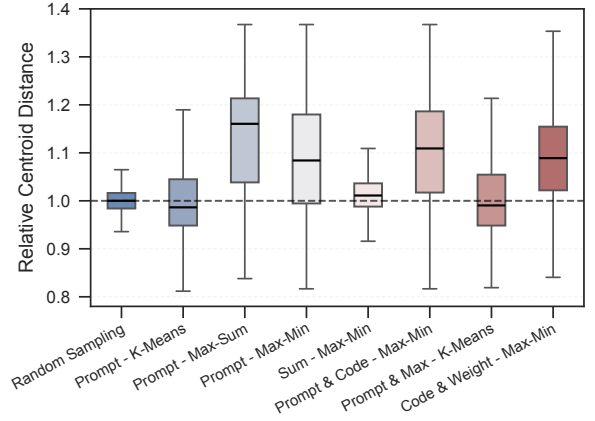


Figure 3: Global dispersion distributions for representative selectors. The dashed line marks $RCD = 1$. For compact labels, *Prompt* and *Code* denote the representation used for subset selection, while *Sum*, *Max*, and *Weight* denote the corresponding macro-intent representations.

Table 3: K-Means normalized mutual information (NMI) between representation-induced clusterings.

K-Means NMI	Micro		Macro	
	Prompt (Qwen3)	Label-Sum	Label-Max	Label-Weight
Prompt (Qwen3)	—	0.3662	0.3621	0.3595
Code (Qwen3)	0.6477	0.3375	0.3325	0.3310
Code (Jina)	0.5587	0.3581	0.3533	0.3509

Overall, these findings support RQ2: semantic anchoring changes the regions selected by standard diversity objectives, reducing the tendency to select locally isolated or globally peripheral samples while preserving non-random exploration. More broadly, they support the geometric-correction view of SAGC: the same diversity objective can behave differently once the representation geometry exposed to it is semantically anchored.

6.4 Anchor Complementarity Analysis

The main results suggest that macro-intent is particularly useful when used as an anchor for dense micro-context. This subsection addresses RQ3 by testing whether macro-intent provides structural information that is not redundant with dense micro-context. We compare representation-induced structure at two levels: NMI between K-Means assignments with $K = 256$ measures global cluster-level agreement, while k NN overlap with $k = 20$ measures local neighborhood agreement. The k NN overlap is the average fraction of shared nearest neighbors across the two spaces. For both metrics, higher values indicate stronger structural agreement.

Tables 3 and 4 show that **macro-intent captures representation structure that is not redundant with prompt or code embeddings.** Across both global and local metrics, dense micro-context views are much more aligned with each other than with macro-intent representations. Prompt-code agreement reaches 0.6477 NMI and 0.5829 k NN overlap for Qwen3, and remains high when

Table 4: k NN overlap between representation spaces.

k NN overlap	Micro	Macro		
	Prompt (Qwen3)	Label-Sum	Label-Max	Label-Weight
Prompt (Qwen3)	–	0.2280	0.2279	0.2170
Code (Qwen3)	0.5829	0.2025	0.2025	0.1927
Code (Jina)	0.5701	0.2030	0.2030	0.1932

using the Jina code encoder. In contrast, agreement between dense micro-context and macro-intent representations is consistently lower, with NMI around 0.33–0.37 and k NN overlap around 0.19–0.23 across the three label aggregations. These global and local gaps reveal a consistent separation between micro–micro and micro–macro representation structure.

This structural separation supports the view of anchor complementarity in RQ3: macro-intent acts as a complementary anchor for dense micro-context rather than a coarse replacement for dense representations. It also helps explain why semantic anchoring differs from generic feature concatenation. Dense micro-views already share similar cluster and neighborhood structure, so adding another dense view is less likely to substantially reshape the geometry exposed to the selector. Macro-intent, by contrast, introduces a different organization into the same representation space, providing a plausible mechanism for why anchoring dense micro-context with macro-intent is more effective than generic feature concatenation.

6.5 Out-of-Benchmark Generalization

This subsection completes RQ1 by evaluating whether the data-efficiency gains observed on LiveCodeBench transfer to additional code-generation benchmarks. Beyond the main LiveCodeBench evaluation, we evaluate a representative subset of methods on HumanEval-Pro and MBPP-Pro from CodeEval-Pro [41], and on the C++ split of Ag-LiveCodeBench-X [3]. CodeEval-Pro emphasizes more complex function-level instruction following and multi-step logic, while Ag-LiveCodeBench-X C++ evaluates performance in a non-Python programming language. We report MCS on all three additional benchmarks, but unlike the main LiveCodeBench evaluation, each checkpoint is evaluated with a single generation run. For compactness, Table 5 uses HE-Pro for HumanEval-Pro and AgLCB-C++ for the C++ split of Ag-LiveCodeBench-X, and writes each method as representation / objective. Additional benchmark details are provided in the repository appendix.

Table 5 shows that **semantic anchoring remains competitive beyond LiveCodeBench, although the gains vary across benchmarks.** On HE-Pro, the benefit is most pronounced: *Prompt & Label-Max / Max-Min* achieves 0.7134, outperforming random sampling (0.6562), the best dense-only result (0.6783), and the best macro-only result (0.6833). On AgLCB-C++, the same configuration obtains the highest score of 0.3097, with a smaller but still positive margin over random sampling and the strongest macro-only variant. On MBPP-Pro, all methods fall within a narrow range, but a SAGC variant, *Prompt & Label-Weight / K-Means*, still achieves the best score of 0.6783.

Overall, these results support the generalization part of RQ1: semantically anchored selection remains competitive

Table 5: Out-of-benchmark generalization results.

Method	HE-Pro	MBPP-Pro	AgLCB-C++
Random Sampling	0.6562	0.6749	0.2942
Prompt / K-Means	0.6540	0.6665	0.2966
Prompt / Max-Min	0.6783	0.6572	0.2963
Label-Max / K-Means	0.6679	0.6652	0.2864
Label-Max / Max-Min	0.6699	0.6736	0.2905
Label-Weight / K-Means	0.6833	0.6573	0.2909
Label-Weight / Max-Min	0.6786	0.6750	0.3052
Prompt & Label-Max / K-Means	0.6656	0.6730	0.2949
Prompt & Label-Max / Max-Min	0.7134	0.6775	0.3097
Prompt & Label-Weight / K-Means	0.6659	0.6783	0.2910
Prompt & Label-Weight / Max-Min	0.6845	0.6757	0.3064

beyond the main benchmark, with the highest observed score on each additional benchmark obtained by a SAGC variant. The competitive macro-only results, especially *Label-Weight / Max-Min* on AgLCB-C++, suggest that macro-intent carries transferable functional information, while the strongest observed results arise when this structure is used to anchor dense micro-context.

7 Conclusion

Diversity-driven data selection is scalable, but its effectiveness depends on whether the representation geometry exposed to the selector reflects training-relevant variation. In code instruction tuning, we diagnose a geometric bias in dense-space selection: standard diversity objectives can over-explore locally isolated or globally peripheral samples without consistently improving downstream utility. To address this issue, we propose Semantically Anchored Geometric Coverage (SAGC), a representation-level correction that anchors dense micro-context with macro-intent labels while keeping standard diversity objectives unchanged. Our experiments show that SAGC enables data-efficient code SFT, remains competitive beyond the main benchmark, changes the regions selected by standard objectives, and benefits from the complementary structure provided by macro-intent anchors.

This study suggests several natural extensions. First, SAGC is designed as a representation-level correction for diversity-driven selection and is complementary to utility-aware signals such as quality, difficulty, or influence; integrating these signals with semantic anchoring is a promising direction. Second, our implementation constructs macro-intent anchors from LLM-generated labels, and future work may explore more robust or lightweight anchor construction. Finally, our experiments focus on a fixed 20k selection budget and a single SFT base model in code instruction tuning, without repeated SFT runs to quantify training-level variance. Extending the evaluation across broader models, budgets, and code-generation settings would further test the generality and stability of semantic anchoring.

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GenAI Usage Disclosure

Generative AI was used in the research pipeline for instruction filtering and response regeneration (Qwen3-32B) and macro-intent label generation (Qwen3-8B), and was also used to assist with manuscript writing and experimental code development. GenAI-assisted manuscript text and code were reviewed by the authors, who take full responsibility for the research, manuscript, code, and reported results.

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